

AE460B Spacecraft Design

TM-03: Communication Subsystem (COM) Design

Manifest Destiny

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This technical memorandum presents the preliminary design and analysis of the Communication Subsystem (COM) for the Solar Gravitational Lens (SGL) spacecraft operating between 650 and 950 AU. Due to extreme free-space path loss at these heliocentric distances, conventional radio-frequency architectures are infeasible; therefore, an infrared optical communication system based on NASA's Laser Communications Relay Demonstration (LCRD) is selected. The system employs a 15 W 1064 nm laser transmitter, a 1 m spacecraft optical aperture, and 30 m class Earth-based ground telescopes. A complete downlink link budget is developed for the worst-case geometry at 950 AU, including free-space path loss, pointing error, optical inefficiencies, and atmospheric attenuation. Using a photon-efficient Pulse Position Modulation (PPM) architecture and a receiver noise bandwidth of 450 kHz, the achieved carrier-to-noise ratio yields a link margin of 3.02 dB, satisfying the mission requirement of greater than 3 dB margin at end-of-life range. System-level interfaces with ADCS, EPS, and thermal subsystems are evaluated, highlighting the tight pointing, power, and heat-rejection constraints imposed by microradian beam divergence and long-duration transmission events. The resulting architecture demonstrates technical feasibility for sustained deep-space optical communications at distances approaching 950 AU.

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I. Mission and Communication Requirements

A. Mission Scenario

The Solar Gravitational Lens will depart from geostationary Earth and take a Hohmann-type transfer to Jupiter and perform a Jupiter gravity assist using an Oberth maneuver at periapsis, then it will in-fall to the solar perihelion at 0.046 AU before a hyperbolic cruise to 650 AU and beyond along the line-of-sight antipodal to Proxima Centauri b. The mission will have an operational science phase between 650 AU to 900 AU before communications are no longer required. A total of three ground stations will keep constant 24/7 line of sight with margin to account for weather. Primary stations will be at Goldstone California, Madrid Spain, and Canberra Australia, with a backup station at Haleakala in Maui, Hawaii. The operations control center will be the Deep Space Operations Center located at the JPL facilities in Pasadena, California. The communications system will require a science data rate of 16 Gbits/year at 900 AU in order to send all necessary data back to Earth.

B. Data Products and Required Links

Required communications for downlink include telemetry such as health data and payload data such as raw data gathered by the telescope. One critical point of health data will be making sure the spacecraft's clocks are aligned with Earth's. Uplink communications will include telecommanding and acknowledgements. Telecommand can tell the spacecraft to realign its clocks, and acknowledgements will allow the spacecraft to confirm its autonomously calculated values with Earth-predictions. Tracking will be performed by adjusting the spacecraft's pitch and yaw to bring the uplink beacon spot to a 2x2 pixel tracking subwindow in the center of the photon counting camera within the telescope. The spacecraft's point-ahead vector is used to control the downlink beam direction so that it is pointing towards the ground stations.[1]

C. Operational Modes

Nominal communications will be the default mode during the spacecraft's transfer trajectory to 650 AU. This will also be the mode whenever the spacecraft isn't actively gathering or transmitting science data. This is a low-rate housekeeping mode during which the system will maintain a 24/7 data link for tracking and the ability to transmit in high-rate for scheduled tasks like clock realignment.

D. Key Requirements

The spacecraft shall support a science data rate of 16 Gbits/year from 900 AU.

The spacecraft shall maintain down-link data rates with Earth at > 100 bps from distances up to 950 AU.

The ground stations shall maintain up-link data rates that support > 50 bps at 950 AU.

The communications system shall maintain 24/7 connection with the spacecraft.

The spacecraft shall have a link margin of greater than 3 dB.

II. COM Architecture

A. Band and Frequency Choice

For traditional, near earth spacecraft, Radio Frequency (RF) is the most common means of telecommunications by far. Many satellites around earth and the moon use Ka-band RF, between 26.5 - 50 GHz. Earlier missions, such as Mars rovers and Voyager 1 utilize X-band RF, at 8 - 12 GHz. X-band is utilized for these missions due to its ability to pass through the atmosphere with minimal interference. Despite this, a mission up to 900 AU into deep space results in a whole other problem. Because data rates scale with an inverse-square of distance, the distance for the SGL becomes an issue that cannot be solved by RF transmissions. Developments in laser communications at NASA has recently become the forefront of deep space data transfers, and the development of the Laser Communications Relay (LCR) demonstrator is the ideal system we will use to communicate with the SGL [2]. Working in the Infrared spectrum (300 THz (3×10^{11} Hz) to 400–430 THz (4.3×10^{14} Hz)), this operates far above the Extremely High Frequency (EHF). The benefit of this frequency is the ability to communicate far more information in a single down link. This means more power can be placed in the range and accuracy, and is able to deliver an even greater data volume. Current testing on the LCR shows it's ability to operate up to 100 times more powerful than modern RF systems.

B. Hardware Block Diagram

The ground station for the LCR included an integrated optical system that can handle pointing regardless of the sun and SGL position relative to earth using a sun shutter. The a parabolic mirror aligns the light, which is fed into the downlink, or to the target from the uplink. A wavefront sensor measures aberrations in the downlink, and a scoring camera records the quality of the feedback. A waveplate is used as the polarizer to filter out sun light, the remaining light is fed into the receiver [2].

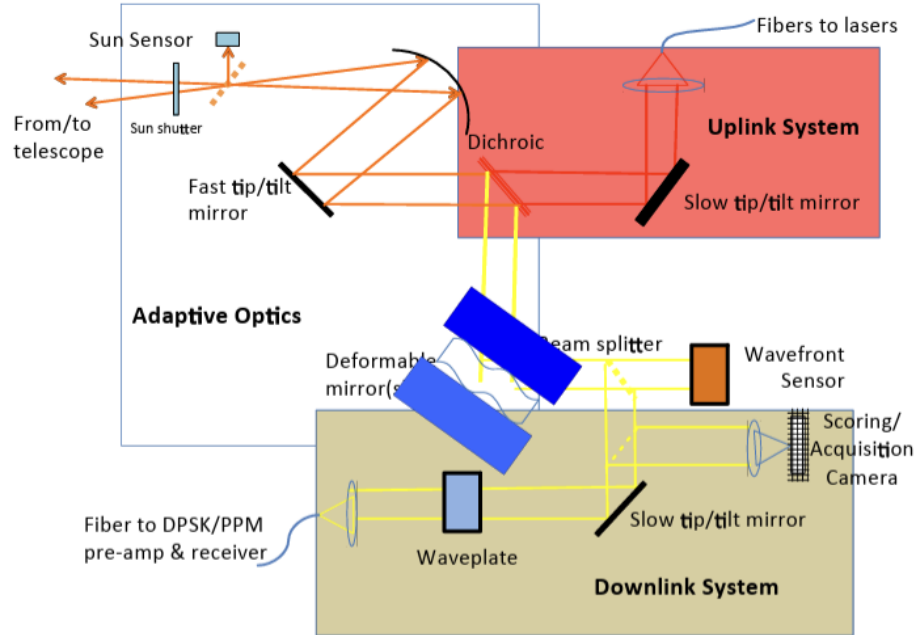


Fig. 1 JPL Optical Laser Telescope Ground Station Diagram

The spacecraft side of the communication system begins with the telescope receiving IR light from the telescope, which is then processed using beam control, detection, and is then filtered away from light used on the payload. Phototransistor are used to convert the IR light into digital information. This data is processed by the On Board Computer (OBC), and after information is consolidated into what will be transmitted back to earth, the data is converted back to IR, amplified, and send as downlink. The path to the ground stations would accumulate error through space and the atmosphere, which will be accounted for during data processing. [3].

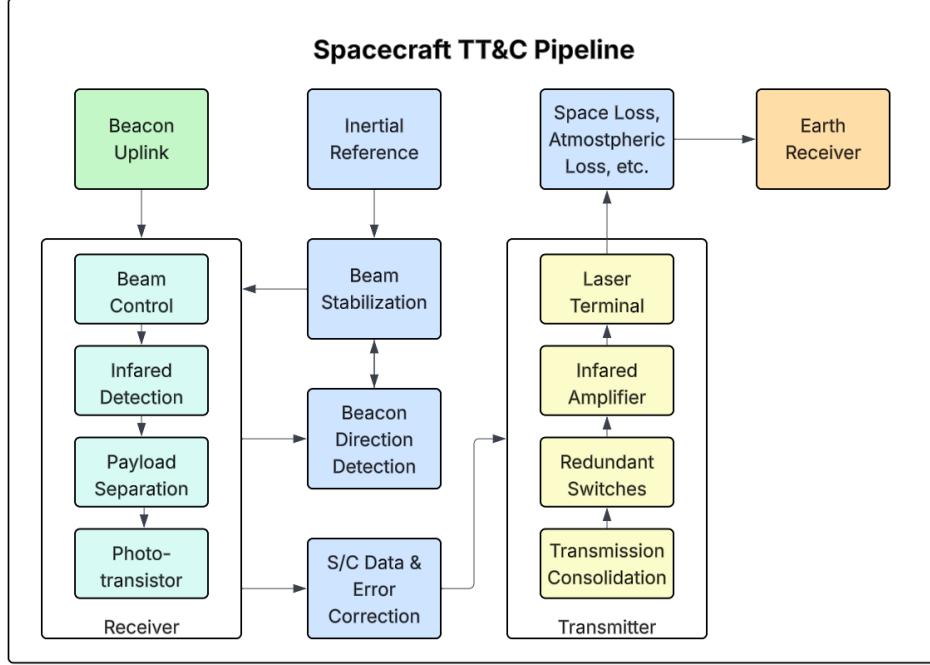


Fig. 2 SGL TT&C Diagram for Laser Communication

C. Point and Coverage Concept

Because we will be receiving so little from the spacecraft, and with such a small target for pointing and receiving, it is important for us to have near constant, clear coverage from earth. In order to have 24/7 clear coverage with the spacecraft, we need a minimum of three ground stations, placed 120° apart around the globe. JPL's Deep Space Network (DSN) has a site at Goldstone California, Madrid Spain, and Canberra Australia. These three sites are already in place and would be prime candidates for the SGL mission [4]. One caveat is the frequency of TT&C. These three sites are appropriate for Voyager I due to the use of X band RF signals, which are highly resistant to atmospheric intrusions, but with IR laser communications, we must rely on good weather in order to receive the already weak signal from days of travel. Because of this, the ideal setup for this project would be to have a minimum of five ground stations, especially ones placed in areas that do not have common weather intrusions. Another primary candidate for this would be Haleakala in Maui, Hawaii. This observatory is in a clean environment without pollution, and rarely has weather intrusions due to its altitude atop a major volcano [5].

III. Link Budget Methodology

A. Geometry and Range

We will use the worst case geometry for our calculations, which will occur at 950 AU, so we can define the following known values.

$$r = 950 \text{ AU, or } 1.4212 \times 10^{11} \text{ km}$$

$$f_t = 2.8176 \times 10^5 \text{ GHz}$$

B. Free-space Path Loss

$$L_{FS} = 20 \log_{10}(f_t) + 20 \log_{10}(r) + 92.44 = 424.4906 \text{ dB} \quad (1)$$

C. Antenna Gains

$$\eta_A = 0.7 \quad d_t = 1 \text{ m}$$

$$G_t \approx 10 \log_{10}(109.66 f_t^2 d_t^2 \eta_A) \approx 127.85 \text{ dB} \quad (2)$$

D. EIRP

$$p_t = 15 \text{ W}$$

$$P_t = 10 \log_{10}(p_t) = 11.7609 \text{ dBW} \quad (3)$$

$$EIRP = P_t + G_t = 139.61 \text{ dBW} \quad (4)$$

E. Receive Power at the Receiver Antenna

$$d_r = 30 \text{ m}$$

$$f_r = 1.9341 \text{e5 GHz}$$

$$\eta_B = 0.7$$

$$G_r \approx 10 \log_{10}(109.66 f_r^2 d_r^2 \eta_B) \approx 154.1237 \text{ dB} \quad (5)$$

$$PointingLoss = 0.8 \text{ dB}$$

$$OpticsLoss = 2 \text{ dB}$$

$$AtmosphericLoss = 2.5 \text{ dB}$$

$$L_{\text{other}} = PointingLoss + OpticsLoss + AtmosphericLoss = 5.3 \text{ dB} \quad (6)$$

$$P_r = EIRP + G_r - L_{FS} - L_{\text{other}} = -136.06 \text{ dBW} \quad (7)$$

F. Noise and Carrier-to-Noise

$$k = 1.380649 \text{e-23 J/K}$$

$$T_s = 500 \text{ K}$$

$B = 450000 \text{ Hz}$ The receiver noise bandwidth is assumed to be 450 kHz based on pulse shaping and matched filtering at the detector front-end.

$$n = 10 \log_{10}(k T_s B) = -145.0773 \text{ dBW} \quad (8)$$

$$\left(\frac{G}{T} \right) = G_r - 10 \log_{10}(T_s) = 127.1340 \text{ dB} \quad (9)$$

$$\left(\frac{C}{N} \right)_{\text{dB}} = EIRP + \left(\frac{G}{T} \right) - L_{FS} - L_{\text{other}} - 10 \log_{10}(k) - 10 \log_{10}(B) = 9.0204 \text{ dB} \quad (10)$$

G. Link Margin Check

$$\left(\frac{C}{N} \right)_{\text{req}} = 6 \text{ dB}$$

$$M_{\text{link}} = \left(\frac{C}{N} \right)_{\text{dB}} - \left(\frac{C}{N} \right)_{\text{req}} = 3.0204 \text{ dB} \quad (11)$$

H. Link Budget Table

Term	Symbol	Value (dB)
Transmit Power	P_t	11.761
Tx Antenna Gain	G_t	127.85
EIRP	$P_t + G_t$	139.61
Free-space Path Loss	L_{FS}	424.49
Other losses (Pointing, Atmospheric, Optics)	L_{other}	5.3
Rx Antenna Gain	G_r	154.12
Received Carrier Power	P_r	-136.06
Noise Bandwidth	B	45000
System Temperature	T_s	500
G/T	$G_r - 10 \log T_s$	127.13
Achieved Carrier to Noise Ratio	C/N	9.2779
Link Margin	M_{link}	3.0204

Table 1 Preliminary Downlink Budget at 950 AU

IV. Antenna Selection and Coverage

A. Decoder Architecture

Since the Solar Gravitational Lens mission relies on optical communications for the primary Earth downlink from 650–950 AU, the traditional radio frequency (RF) "antenna type" will be replaced by a selection of optical modulation and decoder architectures that will enable the spacecraft to return us more reliable data in deep space, where photons will be extremely limited. At these distances, the optical power we receive will be very low, so we must combine photon-counting detection with Pulse Position Modulation (PPM) to achieve higher photon efficiency. Studies made by NASA on optical communications show that PPM is one of the most appropriate modulation schemes for deep-space missions like the SGL [2]. Spacecraft that operate near the Earth typically use Differential Phase Shift Keying (DPSK) instead because it supports higher data rates when sufficient photons are available. The sustained downlink rates of our mission will be on the order of $\gtrsim 100$ bps from about 900 AU. This should be feasible when it is paired with a high-power (15 W) laser transmitter and meter-class optical aperture.

Our decoder architecture must also support photon-counting inputs and forward error corrections while operating within the power and thermal constraints of the spacecraft. The Real Time Optical Receiver (RealTOR) project by NASA has developed a field-programmable gate array (FPGA) that uses parallel decoder slices to increase throughput and shift-register serialization to accommodate higher-order PPM formats without redesigning the core decoding logic. Following this approach shows that we can achieve a good level of scalability through digital parallelism rather than fundamentally changing any coding theory. The spacecraft optical terminal of the SGL will also employ a photon-counting detector that will be coupled with a PPM demodulator and an iterative FED decoder, with both using the FPGA architecture. Unlike RF antenna selection, which primarily relies on geometry and gain, the feasibility of our optical communication system at 900 AU will be closely tied to decoding efficiency, processing capability, and long-term reliability.

B. Beamwidth and Pointing

Conventional RF reflector systems are different from optical communications systems because the beamwidths that they produce are on the order of fractions of a degree to several degrees. Optical communication systems, however, produce extremely narrow beams in the microradian range. For example, NASA's Laser Communications Relay Demonstration (LCRD) optical module demonstrated a 15 microradian downlink beam from a 4-inch telescope [6]. For the SGL mission at a distance of about 900 AU, we must have a very narrow divergence to concentrate optical power toward Earth. The only downside to this is that it fundamentally changes the pointing and stabilization requirements

compared to the RF architecture. Getting modest pointing errors with the RF system can result in small gain degradation, whereas with optical systems, just having a micro-radian level misalignment can significantly reduce the photon flux received, and it can also directly impact our decoder performance and link margin.

Since this narrow beam geometry will be problematic for our mission, our optical terminal will require a layered acquisition and tracking approach rather than using a simple body pointing system. NASA's LCRD demonstrates this because the optical head includes a spatial acquisition detector (e.g. quadrant detector with 2 microradian field of view) that enables an initial signal pull-in. This system is also followed by fine tracking, which uses a two-axis gimbaled telescope with inertial stabilization to reject any kind of disturbances. This architecture is more ideal because it isolates fine pointing control from any jitter the spacecraft bus encounters and allows a continuous alignment of the transmitted beam with the receiver on Earth. Pointing errors must be a small fraction of the beam divergence angle to try to avoid any excessive pointing loss we may deal with, which would ultimately become a dominant contributor to the degradation of our optical link.

An important note to make is that optical links also introduce various sensitivities that are not present in RF systems. These include atmospheric turbulence, scintillation, absorption, and even cloud cover at the ground station that can attenuate or interrupt our link. Conventional RF systems are comparatively more robust to these kinds of effects. These sensitivity issues with optical communications require some site diversity and margin allocation for atmospheric losses. The stability of the beam between the spacecraft and ground station is closely tied to the jitter performance of the ADCS, the structural rigidity of the optical bench, and the thermal stability of the telescope assembly. Having small amounts of thermoelastic distortions or vibration sources can give us misalignment on the scale of microradians. To ensure this does not happen, we must use the optical architecture for our SGL mission so we can achieve high beamwidth and pointing performance, which will directly influence the reliability of our links, subsystem integration, and overall mission feasibility.

V. Mass and Power Budgets

Power and mass estimates were taken from existing spacecraft specifications compiled by NASA.[7]

Component	QTY	Power (W)	Unit Mass (kg)	Total Mass (kg)	Mode
Transceiver	1	15	4.7	4.7	nominal/downlink
Receiver	1	0	300	300	uplink
Total		15		302	

Table 2 Power and Mass of all Components

VI. Interfaces, Risks, and Trades

A. Subsystem Interfaces

For our subsystem interfaces, since our spacecraft operates at extreme heliocentric distances (greater or equal to 550 AU), the communication subsystem (COM) strongly drives the overall spacecraft's architecture. The Interfaces affected here are our ADCS, On board Computer (OBC), EPS/Thermal, and Structures Interface's.

For our ADCS interface the COM subsystem directly drives the spacecraft pointing requirement due to the extremely high antenna gain required for long distance communication. Some Implications we will be dealing with are Fine pointing accuracy needs to be significantly smaller than beam-width, This includes our star tracker and camera. High pointing stability is required, especially during long down-link passes. Jitter from our reaction wheels must be minimized. Finally making sure that for our star tracker readings specifically and initial measurements can be supported at arc-second speeds.

Our on board computer will focus on more data rates and scheduling. We will get this information from our optical instruments and image processing pipelines. These will drive the required data rate, transmitters if used, and scheduling with ground stations. We will need to compress data if possible and we will have the required storage for it.

The Thermal interfaces will focus on mainly power and heat rejection. This will be the largest power consumer on board which will contribute to RF power amplification, modulator, and RF switching. The main implications here are transmission loads, heat dissipation from PA inefficiency, and power sources being affected by peak loads.

Finally the structures interface with impact our COM performance through deployment mechanisms which include our antenna location, how we shift our center of mass, and spacing of electronics. Some key things to consider here are how to decrease signal losses, increases in the inertia of our spacecraft, and how our structures flexibility may introduce oscillations or failure.

B. Design Trades

Some Design trades we looked into for our Solar Gravitational Lens spacecraft are Optical vs RF at 900 AU and Direct-to-earth (DTE) vs Dedicated deep space network architecture. For a SGL operating at 900 AU, the communication architecture is decision that enables our mission itself. There are lots of extreme forces with long ranges which have many trade-offs between performance, cost, complexity, and subsystem impacts.

Trade 1: At 900 AU, our spacecrafts path loss will be enormous which can have major affects on our ability to acquire viable data. To achieve viable data we would need either high antenna gain or very precise optical beams. The advantages of this are the potential for very high data rates, smaller transit area compared to equal RF Gain, and this would mean lower transmit power for equal link margin. However some challenges include very small pointing requirements, extreme sensitivity to spacecraft jitter and structural stability, and operational complexity. From this we can conclude that Optical communication does offer us ,in theory, higher performance but it does significantly increase ADCS and operational risks. RF provides lower technical risk but increases mass and power demands.

Trade 2: A DTE vs Dedicated Deep Space Network Architecture essentially evaluates either if the spacecraft would rely on direct-to-earth communication using established infrastructure (Option A) or whether mission's architecture should include dedicated communication assets onboard (Option B). For Option A we would use the existing Deep Space Network (DSN) which would give us proven global coverage and lower nonrecurring development cost. However the issues are we would have to compete with other ongoing missions and limited availability during peak operational periods. There is option B which is having a custom infrastructure. This would allow us for greater mission autonomy and non-conflict with other missions but this would significantly increase our costs and we could potentially be burdened with long-term maintenance and operations. Overall Direct-to-earth communication via DSN is generally cost-effective and has a lower risk than developing new dedicated infrastructure. Keeping in mind long mission duration may increase constant operational costs.

C. Key Risks

Due to the extreme range and narrow beam-width requirements, very small deviations in performance assumptions can significantly decrease the results we are looking for. When talking about insufficient link margin, specifically for optical systems, the atmospheric turbulence and cloud cover could reduce our link reliability. Therefore for our RF systems we will need to pay close attention to the path we take and how our signal strength differs. Here the main risk would be a reduced data rate and an increased bit rate. This will be mitigated by deep elevation analysis during our early design phases, utilizing more than one ground station, and we will use adaptive coding and modulation to maintain our link integrity.

The risk of our pointing losses being a lot larger than we initially assumed also plays a huge role. With high antenna gain this will directly correspond with the narrow beam-width plus any additional pointing error will cause gain reduction. The impact of this risk is mainly the reduction of antenna gain and data loss. This will be mitigated by the implementation of structural interfaces that are high stiffness and low vibration and maintain the ADCS performance margin beyond our minimum requirements.

The Regulatory/ Frequency systems require lots of internal coordination to avoid interference with any other systems that may be nearby. Our long duration mission will increase the chance of regular changes over our spacecrafts life-cycle. The risk itself comes from delays in frequency and operational on Bandwidth or transmit power restrictions. We will mitigate this by maintaining flexibility in modulation and our bandwidth configuration as well as possibly using deep-space standard frequency bands.

Finally, we consider the thermal limits during downlink. During long mission we know that at high speeds our Power amplifier will dissipate significant heat. This heat rejection really falls upon the radiative cooling we will have in place. We have to be aware that peak downwind sessions may exceed our thermal capacity in a steady state. The main risk here will be the preservation of our transmitters and the amount of time our transmitters will last. Essentially We don't want our hardware to become severely degraded. We will mitigate this by using high-efficiency amplifiers to reduce heat waste and we will closely monitor PA temperature by implementing an autonomous sensor. Overall at these extreme heliocentric distances, communication risk is very closely related to our ADCS performance, power availability,

and thermal rejection capability.

VII. Conclusion

The Solar Gravitational Lens Spacecraft's Communication System will be an optical communication system based on the NASA Laser Communications Relay Demonstrator that will send uplink and downlink using light in the infrared spectrum with uplink at 1550 nm and downlink at 1064 nm. The optical modulation will use Pulse Position Modulation (PPM) while the decoder architecture will utilize a photon-counting detector with a PPM demodulator and iterative FED decoder. Three primary ground stations with one redundant ground station will maintain a 24/7 connection. The link margin at the worst case, 950 AU away from Earth, exceeds the required value with 3.0204 dB. The onboard communications transceiver will be 4.7 kg and use 15 W of power. Assumptions include atmospheric, optical, and pointing losses which can exceed the assumed values. The project also assumes ground stations have been built with 30 m telescopes designed for optical communication. The project assumes that line of sight with Earth isn't broken at any point and other idealizations. Risks are prevalent due to the low Technology Readiness Level (TRL) of this system. Extreme nanoradian pointing accuracy is needed, meaning any slight movement can interrupt connection. The vast distances covered by the spacecraft will cause delays and introduce opportunities for error. Finally, the long mission duration can wear down components. Key risks will hinge upon the attitude control subsystem, electrical-power subsystem, and the thermal subsystem's performance.

Nomenclature

<i>ADCS</i>	=	Attitude Determination and Control System
<i>AU</i>	=	Astronomical Unit ($1.496 * 10^{11}m$)
<i>B</i>	=	Noise Bandwidth
<i>C</i>	=	Carrier Power
<i>COM</i>	=	Communication Subsystem
<i>d_t</i>	=	Diameter of Receiver
<i>DSN</i>	=	Deep Space Network
<i>EIRP</i>	=	Effective Isotropic Radiated Power
<i>f_r</i>	=	Frequency of Output
<i>G</i>	=	Gain
<i>IR</i>	=	Infrared
<i>k</i>	=	Boltzmann Constant
<i>LCR</i>	=	Laser Communications Relay
<i>LCRD</i>	=	Laser Communications Relay Demonstration
<i>N</i>	=	Noise Power
<i>OBC</i>	=	On Board Computer
<i>p_t</i>	=	Transmitting Power (W)
<i>P_t</i>	=	Transmitting Power (dBW)
<i>RF</i>	=	Radio Frequency
<i>SGL</i>	=	Solar Gravitational Lens
<i>TT&C</i>	=	Telemetry Tracking and Command
<i>η</i>	=	Antenna Efficiency

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